

PEIRA: Learning Predictive Encoders through Inter-View Regressor Alignment

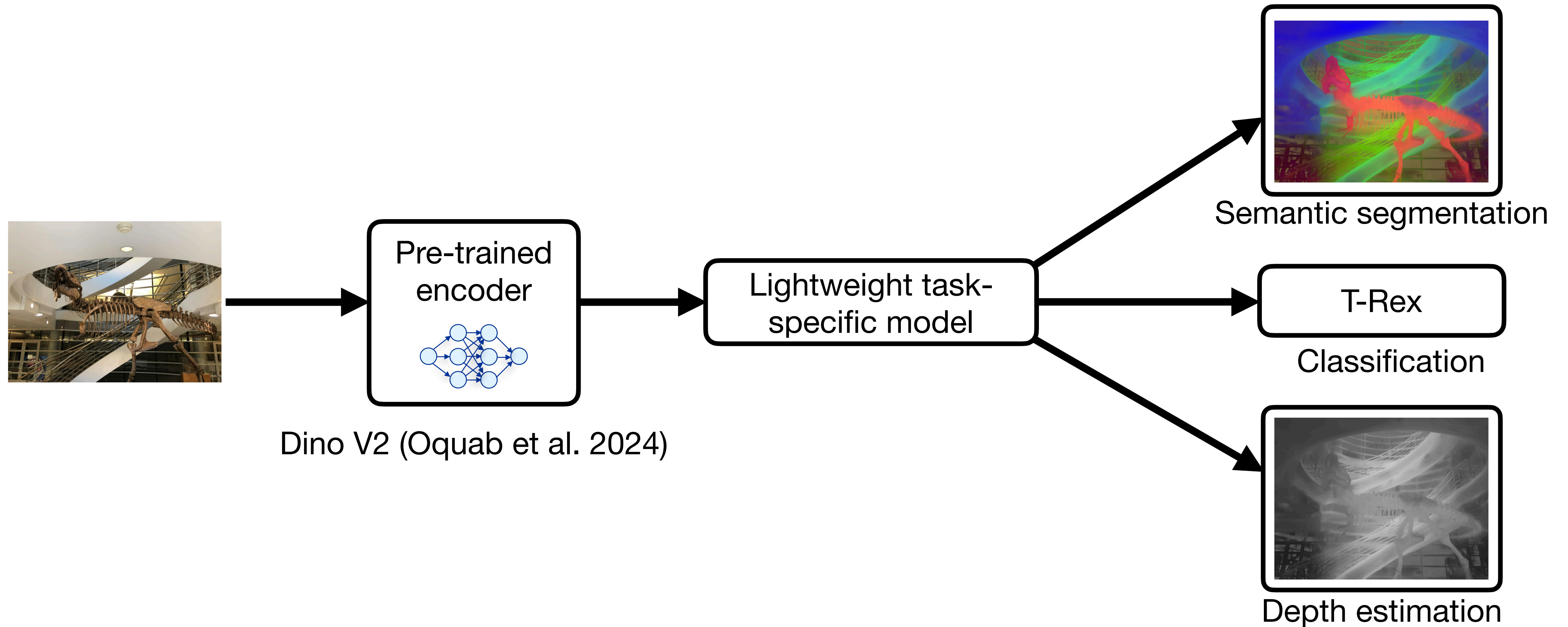
Optimization and Learning Day
GHOST, Inria

Michael Arbel
Inria

June 17, 2026

Context and motivation

Self-supervised pre-training can produce highly transferable visual features

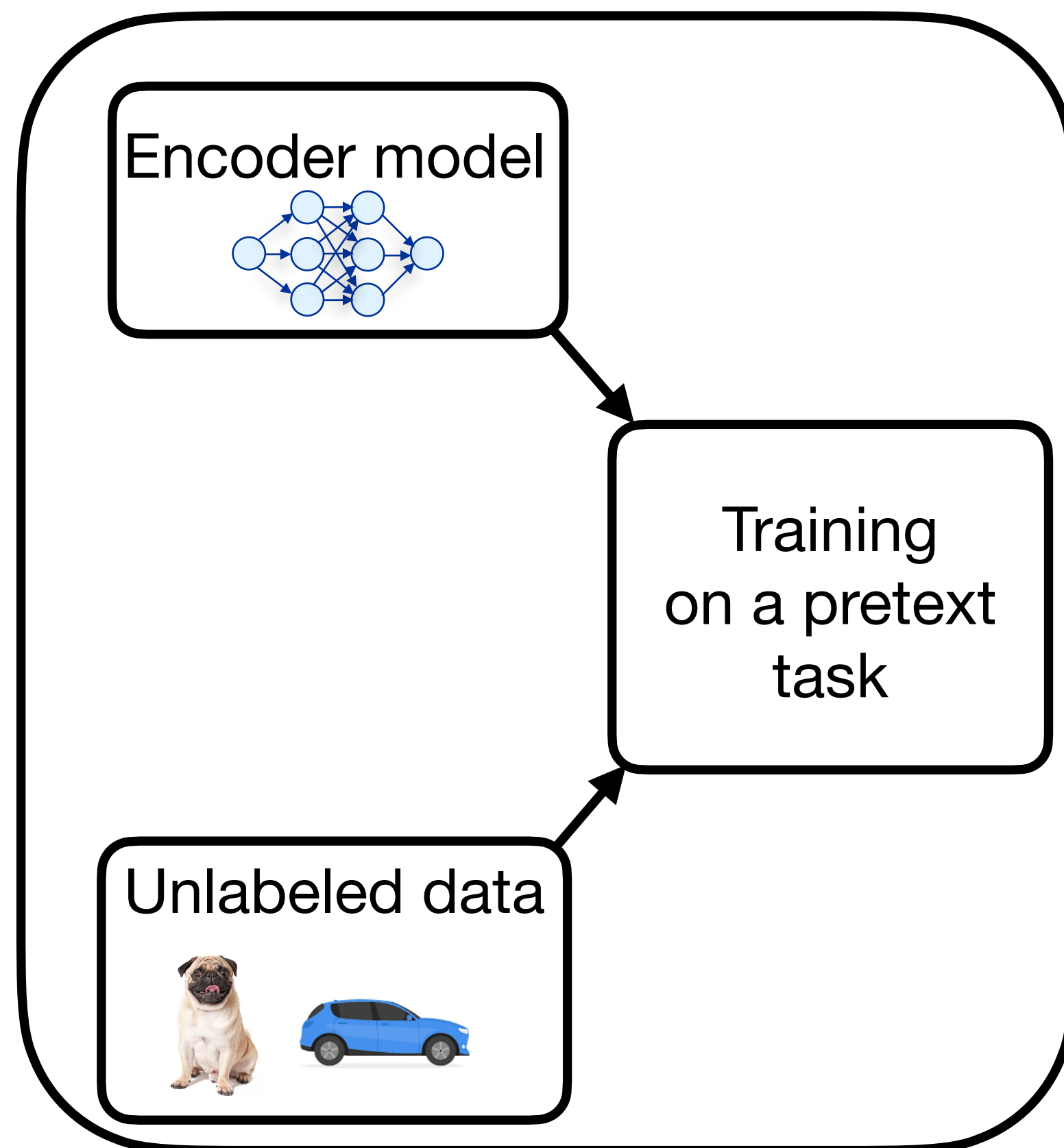


Yet, why these features emerge and what structure they capture is poorly understood

Self-supervised learning (SSL)

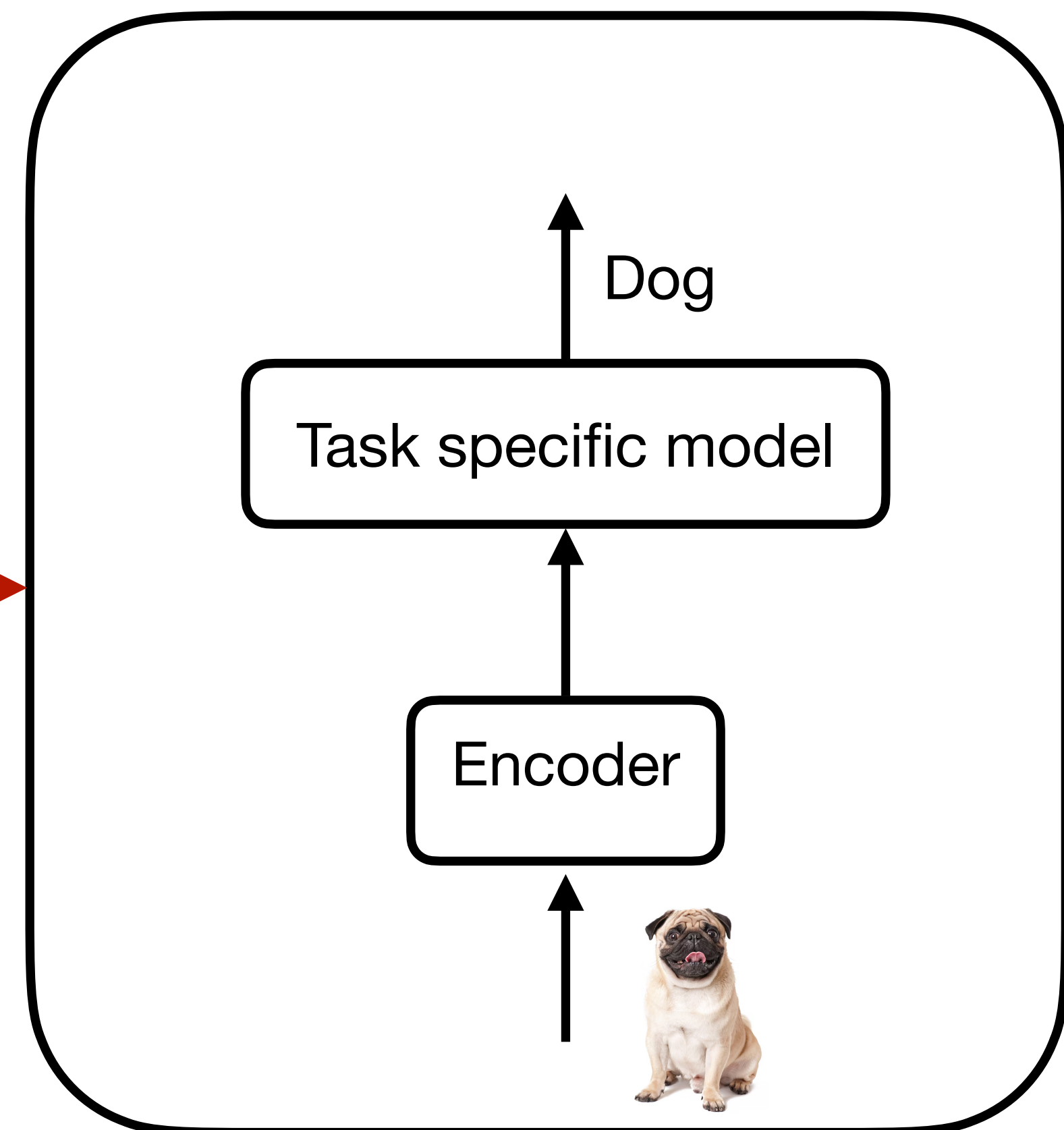
Goal: Learn a useful data representation without supervision

SSL pre-training



Pre-trained
encoder

Downstream task



Self-supervised learning (SSL)

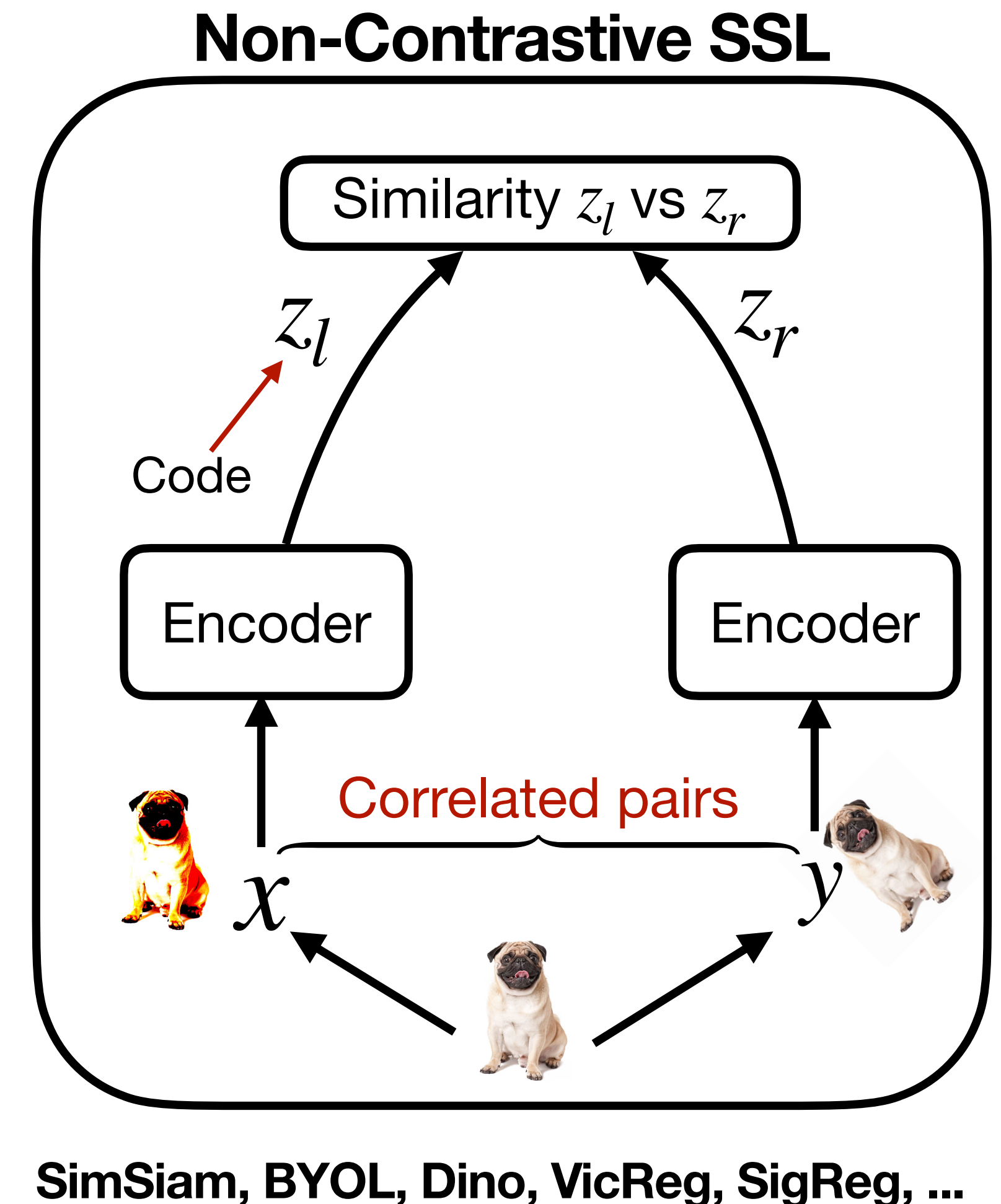
Relies on comparing correlated pairs of data constructed from the same input

Main approaches:

- Reconstruction-based: predict missing/corrupted input
- Contrastive: match positives, separate negatives
- Non-contrastive: predict missing/corrupted representation

Focus: Non-contrastive SSL

- State-of-the-art performance
- Can collapse to trivial representation

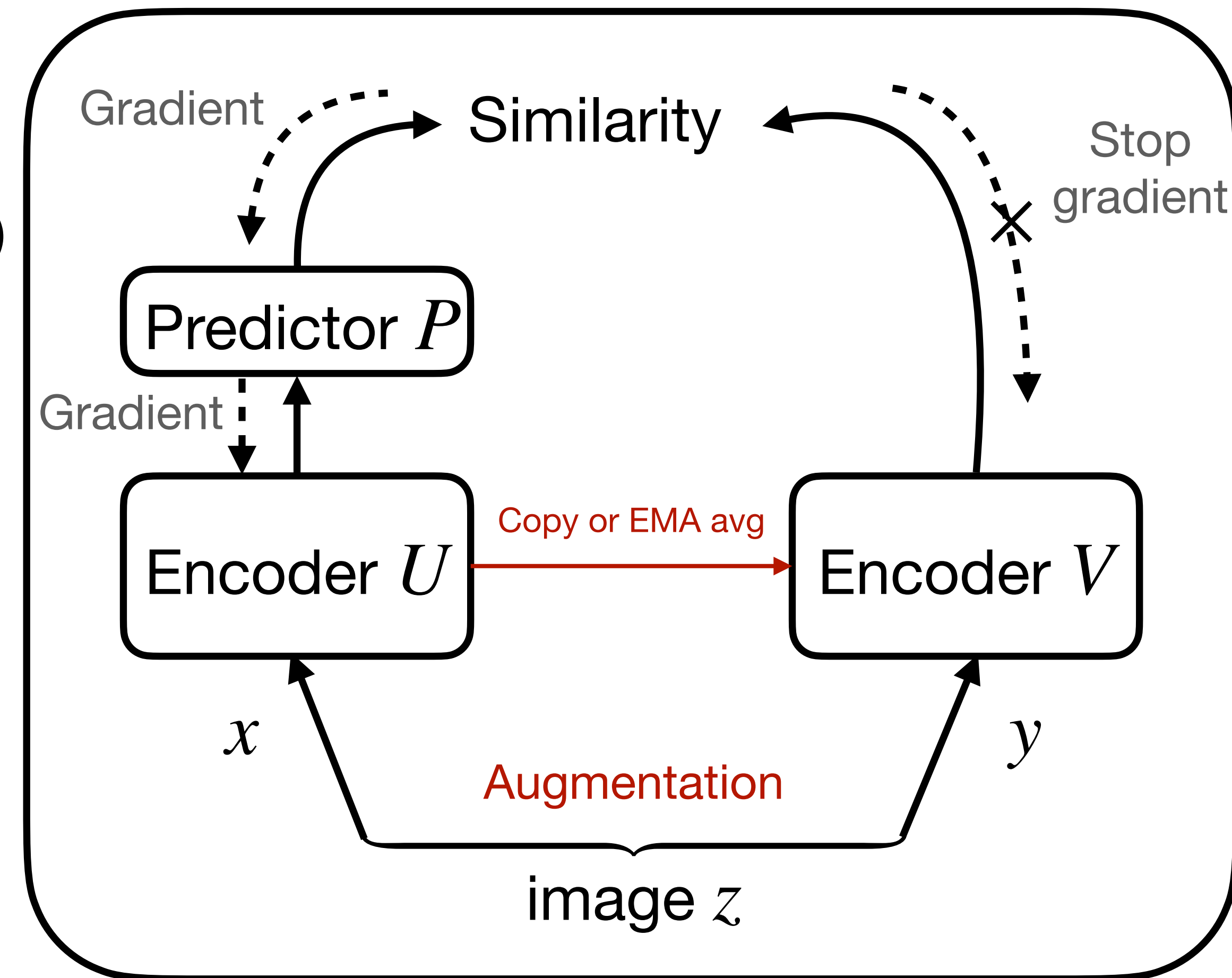


Self-distillation methods for non-contrastive SSL: A puzzle

- Uses heuristics to avoid collapse (Stop-gradient, EMA)
- Not an optimization algorithm [Ponce et al, 2026]
- Excellent empirical performance (e.g., Dino v1, v2, v3)

Goal: Understanding:

- What is learned?
- When does collapse occur?
- How to reliably prevent collapse? Bilevel optimization to the rescue!



Self-distillation SSL: Existing theory is informative but restricted

Special setting

- Linear encoders U, V
- Particular data covariance structure
- Special initialization (ex: orthogonal)

Tian et al. 2021

Littwin et al. 2024

Liu et al. 2022

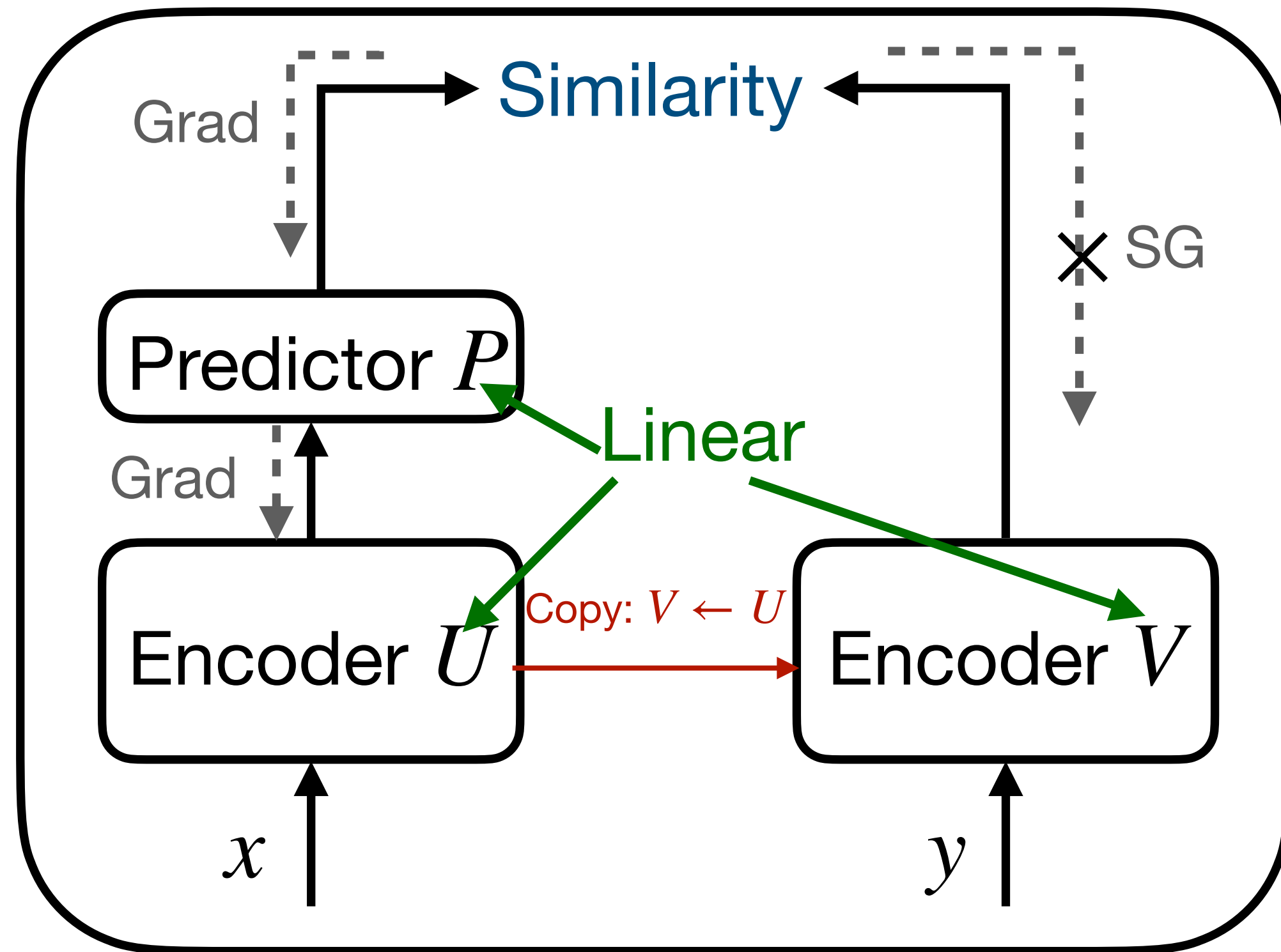
Tang et al. 2023

?

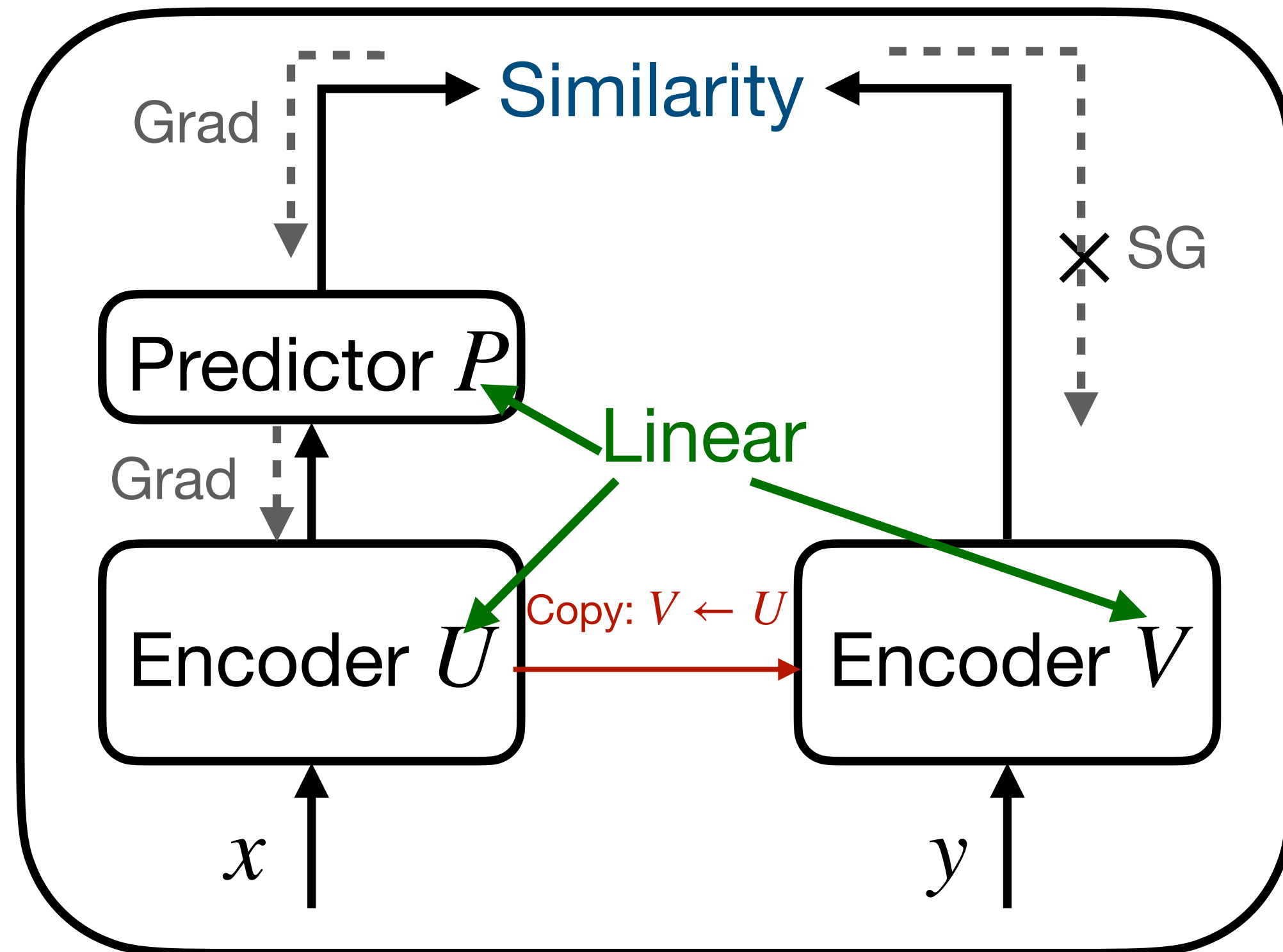
General setting

- Non-linear encoders U, V
- General data distributions
- General initialization

SimSiam-like SSL: Linear encoder and predictor



Self-distillation SSL: Linear encoder and predictor



Similarity

$$\mathcal{L}(P, U, V) = \frac{1}{2} \mathbb{E}[\|PUx - Vy\|^2] + \frac{\lambda}{2} \|P\|^2 + \frac{\lambda}{2} \|U\|^2$$

Continuous-time dynamics

$$\begin{cases} \dot{P} = -\partial_P \mathcal{L}(P, U, V) \\ \dot{U} = -\partial_U \mathcal{L}(P, U, V) \\ V = U \end{cases}$$

Not the gradient flow of any objective
(Ponce et al. 2025)

Linear encoder and optimal predictor (Tang et al. 2023)

$$\mathcal{L}(P, U, V) = \frac{1}{2} \mathbb{E}[\|PUx - Vy\|^2] + \frac{\lambda}{2} \|P\|^2 + \frac{\lambda}{2} \|U\|^2$$

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Optimal predictor

$$\begin{cases} P^\star = \arg \min_P \mathcal{L}(P, U, V) \\ \dot{U} = -\partial_U \mathcal{L}(P^\star, U, V) \\ V = U \end{cases}$$

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Optimal predictor

$$\begin{cases} P^* = \arg \min_P \mathcal{L}(P, U, V) \\ \dot{U} = -\partial_U \mathcal{L}(P^*, U, V) \\ V = U \end{cases}$$

Theorem (Adapted)

Assume:

- Orthogonal initialization $U_0 U_0^\top = I \in \mathbb{R}^{k \times k}$
- Matrix $\Sigma_{xx} = \mathbb{E}[xx^\top] = \alpha I$
- Matrix $\Sigma_{yx} = \mathbb{E}[yx^\top]$ is symmetric

Then:

- U stays orthogonal over time
- $\mathcal{F}(U) = -\frac{1}{2} \text{Tr}((U \Sigma_{yx} U^\top)^2)$ decreases over time
- Rows of an equilibrium U^* span
an eigen-subspace of Σ_{yx} of at most k dimensions

Linear encoder and optimal predictor (Tang et al. 2023)

$$\mathcal{L}(P, U, V) = \frac{1}{2} \mathbb{E}[\|PUx - Vy\|^2] + \frac{\lambda}{2} \|P\|^2 + \frac{\lambda}{2} \|U\|^2$$

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Q: Convergence to an equilibrium U^* ? Which eigen-subspace of Σ_{yx} is selected?

A: Unknown

Linear encoder and optimal predictor (Tang et al. 2023)

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Q: General initialization? General data? (Still with linear encoders)

A: **Complex dynamical system, hard to analyze (Ponce et al. 2025)**

Linear encoder and optimal predictor (Tang et al. 2023)

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Q: What about non-linear encoders?

A: Seems even harder, but not so if done in function spaces

PEIRA: Learning Predictive Encoders through Inter-View Regressor Alignment

Michael Arbel*

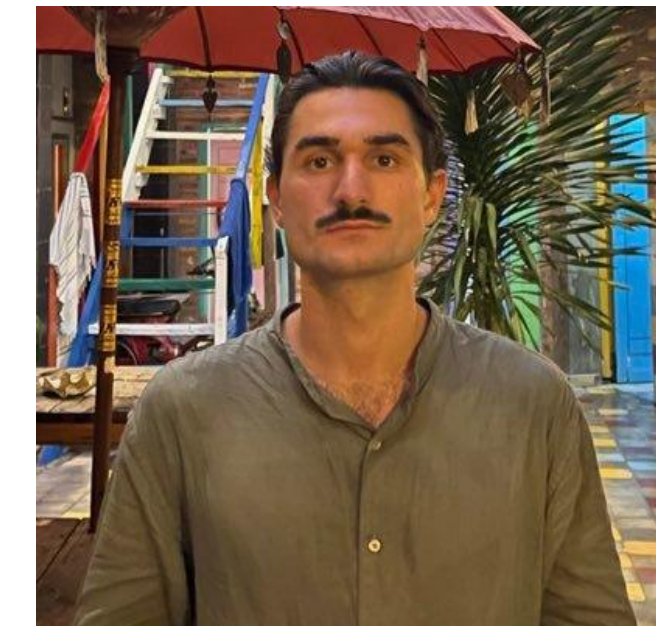
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- Characterization of a non-linear self-distillation dynamics for SSL
 - Limiting objects, convergence of the dynamics and characterization of collapse
- Bilevel optimization alternative to SSL that provably avoids collapse

A self-distillation dynamics in function space

$$\mathcal{L}(P, U, V) = \frac{1}{2} \mathbb{E}[\|PU(x) - V(y)\|^2] + \frac{\lambda}{2} \|P\|^2 + \frac{\lambda}{2} \mathbb{E}[\|U(x)\|^2]$$

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- Optimal linear predictor (as in Tang et al. 2023)
- No parametric form for the encoder U
- Dynamics defined in function space $L_2(\mathbb{P}_x, \mathbb{R}^k)$

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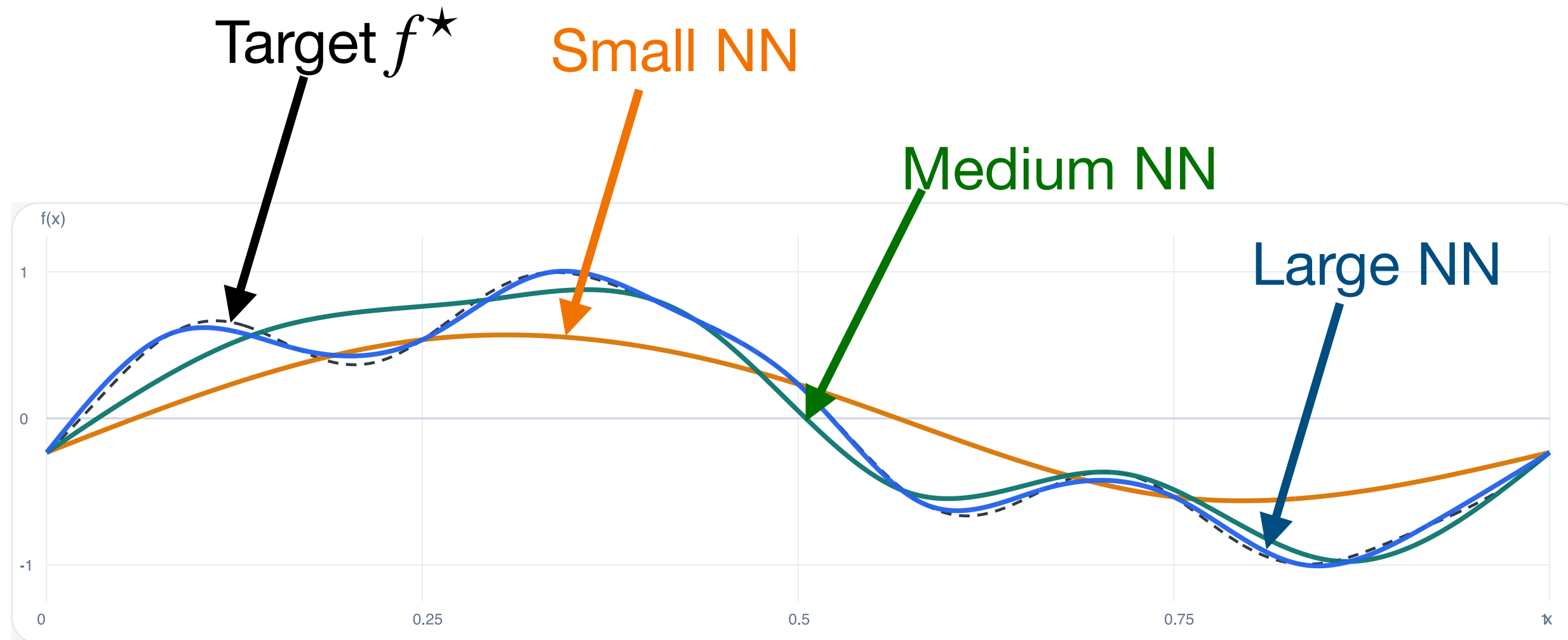
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Why care about function spaces?

Aren't we training a neural network after all?

Why care about function spaces?

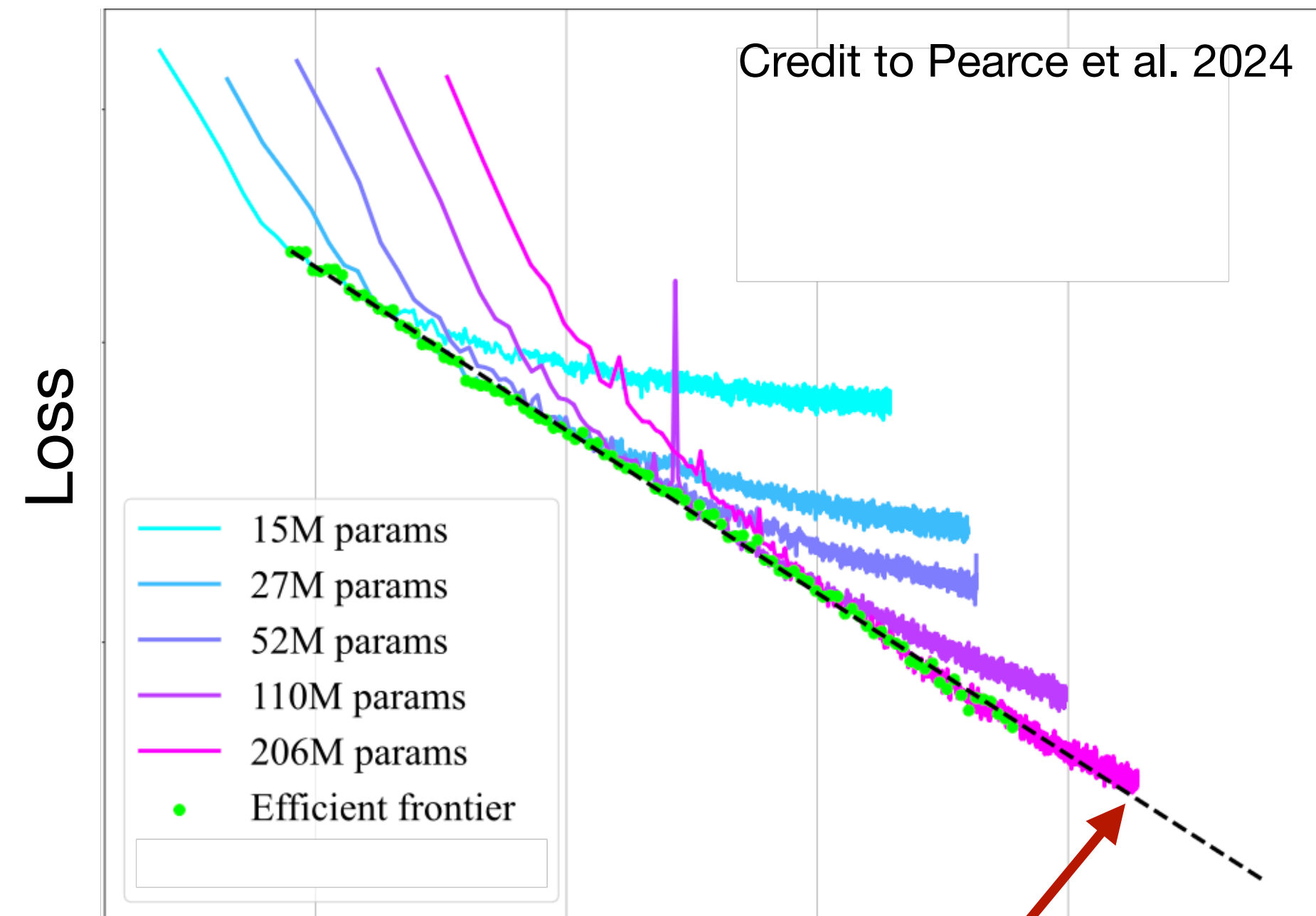
Neural Networks are function approximators



Regression: $f^* = \arg \min_f \mathbb{E} [\|y - f(x)\|^2]$

Target: $f^*(x) = \mathbb{E} [y | x]$

Scale expressivity, data and compute



Approach a functional target

Function space analysis reveals the target (properties, stability, etc)

A self-distillation dynamics in function space

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A self-distillation dynamics in function space: Convergence

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- Optimal linear predictor (as in Tang et al. 2023)
- No parametric form for the encoder U
- Dynamics defined in function space $L_2(\mathbb{P}_x, \mathbb{R}^k)$

Theorem (Informal): Assume x and y play symmetric roles and data regular enough.

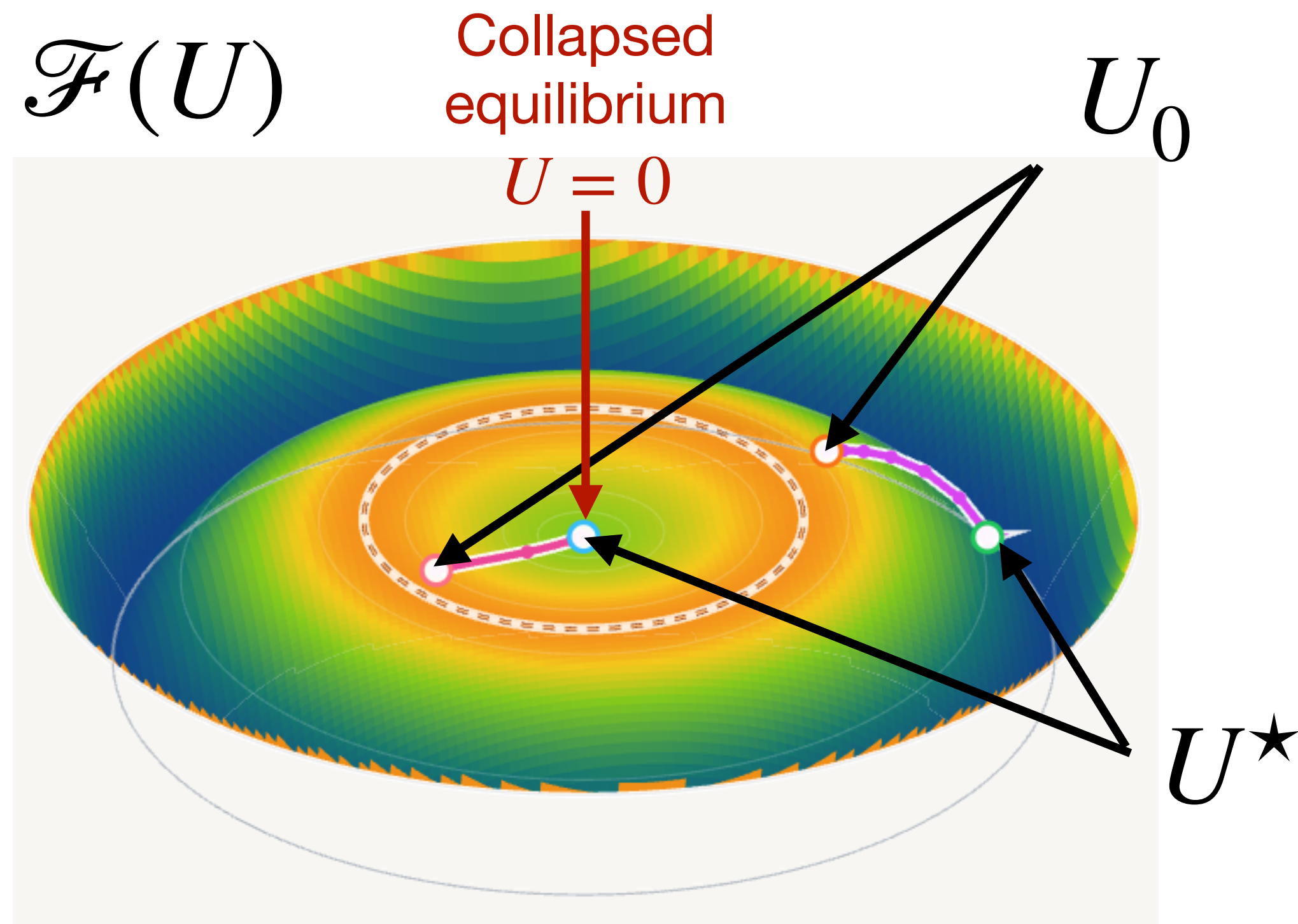
Then: 1- For any initialization U_0 , the dynamics converges to an equilibrium $U^\star$.

2- The following objective decreases in time (Lyapunov function):

$$\mathcal{F}(U) = -\frac{1}{2}\text{Tr}(\mathbb{E}[U(x)U(y)^\top]^2) + \frac{1}{3}\text{Tr}((\mathbb{E}[U(x)U(x)^\top] + \lambda I)^3)$$

No assumption on the data covariances, or orthogonal initialization, as in Tang et al. 2023

A self-distillation dynamics in function space: Convergence



- No limit cycles, no divergence.
- Convergence to a critical point of the Lyapunov function.

Key technical ingredients

- **Gradient-like system:** The vector field aligns well with the gradient of the Lyapunov function
- **Lojasiewicz-Simon inequality:** Small gradients $\implies$ Small gap to a critical value.
- **Trapping argument:** The trajectory is trapped near an accumulation point (Bolte et al. 2014)
- **Existence of accumulation points:** Non-trivial in infinite dimensions (Palais & Smale 1964)

A self-distillation dynamics in function space: Equilibria

What are the equilibria? Which ones are stable?

Canonical Correlation Analysis to the rescue

Linear Canonical Correlation Analysis

Correlation maximization

$$\max_{U, V \in \mathbb{R}^{k \times d}} \mathbb{E}[(Ux)^\top (Vy)]$$

s.t. $\mathbb{E}[(Ux)(Ux)^\top] = I$

$$\mathbb{E}[(Vy)(Vy)^\top] = I$$

whitening constraints

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whitening constraints

Whitened cross-covariance

$$M = \Sigma_{yy}^{-\frac{1}{2}} \Sigma_{yx} \Sigma_{xx}^{-\frac{1}{2}}$$

$\mathbb{E}[yy^\top]$ $\mathbb{E}[yx^\top]$ $\mathbb{E}[xx^\top]$

Singular value decomposition

$$M = (v_1, \dots, v_d) \begin{pmatrix} s_1 & & \\ & \ddots & \\ & & s_d \end{pmatrix} \begin{pmatrix} u_1^\top \\ \vdots \\ u_d^\top \end{pmatrix}$$

CCA solution: top-k singular vector

$$U = \Sigma_{xx}^{-\frac{1}{2}} \begin{pmatrix} u_1 \\ \vdots \\ u_k \end{pmatrix}, \quad V = \Sigma_{yy}^{-\frac{1}{2}} \begin{pmatrix} v_1 \\ \vdots \\ v_k \end{pmatrix}$$

Linear Canonical Correlation Analysis

Correlation maximization

$$\max_{U, V \in \mathbb{R}^{k \times d}} \mathbb{E}[(Ux)^\top (Vy)]$$

s.t.
$$\begin{aligned} \mathbb{E}[(Ux)(Ux)^\top] &= I \\ \mathbb{E}[(Vy)(Vy)^\top] &= I \end{aligned}$$

whitening constraints

Recovers similar eigen-subspace as Tang et al. 2023
under their assumptions
(Σ_{yx} symmetric and $\Sigma_{xx} = \Sigma_{yy} = \alpha I$)

Whitened cross-covariance

$$M = \Sigma_{yy}^{-\frac{1}{2}} \Sigma_{yx} \Sigma_{xx}^{-\frac{1}{2}}$$

$\nwarrow \quad \nearrow \quad \nwarrow$
 $\mathbb{E}[yy^\top] \quad \mathbb{E}[yx^\top] \quad \mathbb{E}[xx^\top]$

Singular value decomposition

$$M = (v_1, \dots, v_d) \begin{pmatrix} s_1 & & \\ & \ddots & \\ & & s_d \end{pmatrix} \begin{pmatrix} u_1^\top \\ \vdots \\ u_d^\top \end{pmatrix}$$

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Non-Linear Canonical Correlation Analysis (Michaeli et al. 2015)

Correlation maximization

$$\max_{(U,V) \in \mathcal{U} \times \mathcal{V}} \mathbb{E}[U(x)^\top V(y)]$$

$L_2(\mathbb{P}_x)^k$

$L_2(\mathbb{P}_y)^k$

s.t.

$$\mathbb{E}[U(x)U(x)^\top] = I$$

$$\mathbb{E}[V(y)V(y)^\top] = I$$

whitening constraints

Non-Linear Canonical Correlation Analysis (Michaeli et al. 2015)

Correlation maximization

$$\max_{(U,V) \in \mathcal{U} \times \mathcal{V}} \mathbb{E}[U(x)^\top V(y)]$$

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s.t.

$$\mathbb{E}[U(x)U(x)^\top] = I$$

$$\mathbb{E}[V(y)V(y)^\top] = I$$

whitening constraints

Non-linear CCA solution: top-k canonical functions

$$U(x) = \begin{pmatrix} u_1(x) \\ \vdots \\ u_k(x) \end{pmatrix}, \quad V(y) = \begin{pmatrix} v_1(y) \\ \vdots \\ v_k(y) \end{pmatrix}$$

Canonical correlation $1 = s_1 \geq s_2 \geq \dots$

$$\mathbb{E}[u_i(x)v_i(y)] = s_i \qquad \mathbb{E}[u_i(x)v_j(y)] = 0$$

De-correlation (when $i \neq j$)

Orthonormal basis of $L_2(\mathbb{P}_y)$

Orthonormal basis of $L_2(\mathbb{P}_x)$

SimSiam-like dynamics in function space: Equilibria and Stability

Theorem (Informal):

Equilibria span CCA subspaces. The only stable ones are of the form:

$$U^\star(x) = \sum_{i=1}^r f(s_i, \lambda) q_i u_i(x)$$

Diagram illustrating the components of the equilibrium form:

- Any $0 \leq r \leq k$** : Points to the summation index r .
- Regularization**: Points to the parameter λ .
- Any orthonormal family $(q_i)_{1 \leq i \leq r}$ of $\mathbb{R}^k$** : Points to the vector q_i .
- Canonical function**: Points to the function $u_i(x)$.
- Canonical correlation $s_1 \geq s_2 \geq \dots$** : Points to the scalar s_i .
- Spectral filter**: Points to the function $f(s, \lambda)$.

The spectral filter function is defined as:

$$f(s, \lambda) = \frac{1}{\sqrt{2}}(s + \sqrt{s^2 - 4\lambda}) \mathbf{1}_{s^2 \geq 4\lambda}$$

Collapse occurs when an admissible CCA direction ($f(s_i, \lambda) > 0$ for $i \leq k$) is not selected ($r < i$)

**Preventing collapse through
bilevel optimization**

Intuition and Motivation: Trace maximization of a predictor

- Optimal predictor between two views:

The diagram shows the formula for the optimal predictor $P^\star$ with three annotations:

- Shared signal:** A red arrow points to the matrix $\mathbb{E} [V(y)U(x)^\top]$, which is highlighted in a purple box.
- Within view variability (noise):** A blue arrow points to the matrix $\mathbb{E} [U(x)U(x)^\top]$, which is highlighted in a blue box.
- Signal to noise ratio (matrix form):** A green arrow points to the entire term $\left(\mathbb{E} [U(x)U(x)^\top] + \lambda I \right)^{-1}$, which is highlighted in a green box.

$$P^\star = \mathbb{E} [V(y)U(x)^\top] \left(\mathbb{E} [U(x)U(x)^\top] + \lambda I \right)^{-1}$$

Intuition and Motivation: Trace maximization of a predictor

- Optimal predictor between two views:

The diagram shows the formula for the optimal predictor $P^\star$ with three annotations: a red arrow labeled "Shared signal" pointing to the $\mathbb{E}[V(y)U(x)^T]$ term, a blue arrow labeled "Within view variability (noise)" pointing to the $\mathbb{E}[U(x)U(x)^T]$ term, and a green arrow labeled "Signal to noise ratio (matrix form)" pointing to the entire denominator expression.

$$P^\star = \mathbb{E}[V(y)U(x)^T] \left(\mathbb{E}[U(x)U(x)^T] + \lambda I \right)^{-1}$$

$Tr(P^\star)$ acts as a scalar summary of the matrix signal to noise ratio

Predictive Embeddings through Inter-View Regressor Alignment

- Find optimal symmetrized regressor for each encoders pairs (U, V)

$$\frac{1}{2}\mathbb{E}[\|PU(x) - V(y)\|^2] + \frac{1}{2}\mathbb{E}[\|PV(y) - U(x)\|^2] + \frac{\lambda}{2}\|P\|^2$$

$$P_{U,V}^\star = \arg \min_{P \in \mathbb{R}^{k \times k}} \mathcal{L}_{reg}(P; U, V)$$

- Learn encoders by minimizing regularized negative trace of optimal regressor:

Negative trace minimization to align representation

Regularization to fix the scale

$$\mathcal{L}_{PEIRA}(U, V) = -\frac{1}{2}\text{Tr}(P_{U,V}^\star) + \frac{\lambda}{2}\mathbb{E}[\|V(x)\|^2 + \|U(y)\|^2]$$

PEIRA objective: Global maximizers

Theorem (Informal): Assume the data distribution is regular enough (existence of non-linear CCA).

Then: 1- Global minimizers of $\mathcal{L}_{PEIRA}$ are exactly:

$$(U^\star(x), V^\star(y)) = \sum_{i=1}^k f(s_i, \lambda) q_i (u_i(x), v_i(y))$$

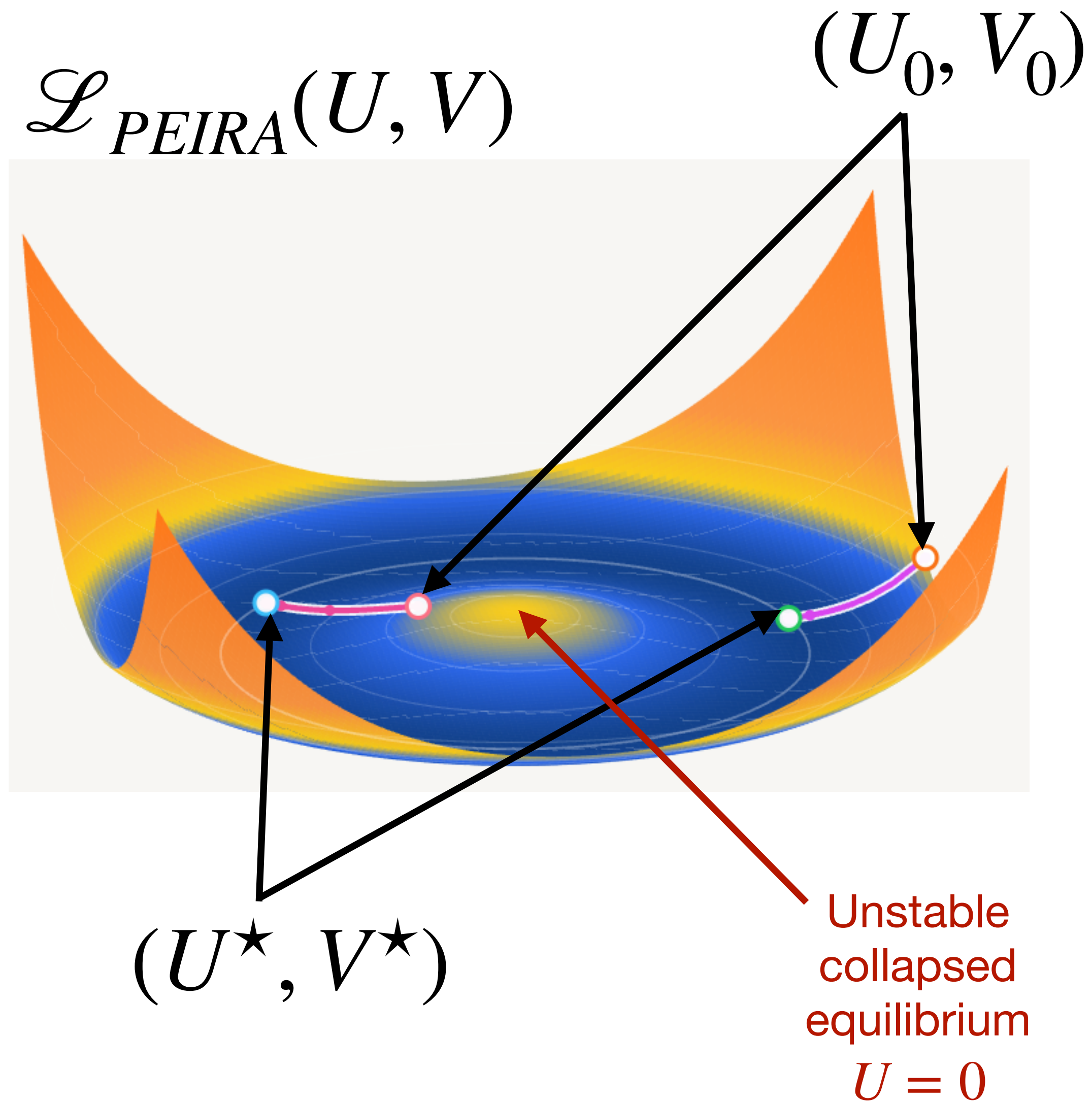
Encoder's dimension $\rightarrow k$

Any orthonormal family $(q_i)_{1 \leq i \leq r}$ of $\mathbb{R}^k$

Canonical functions $(u_i(x), v_i(y))$

Soft-thresholding filter $\frac{1}{\sqrt{2}} \left(\max(\sqrt{s} - \lambda), 0 \right)^{\frac{1}{2}}$

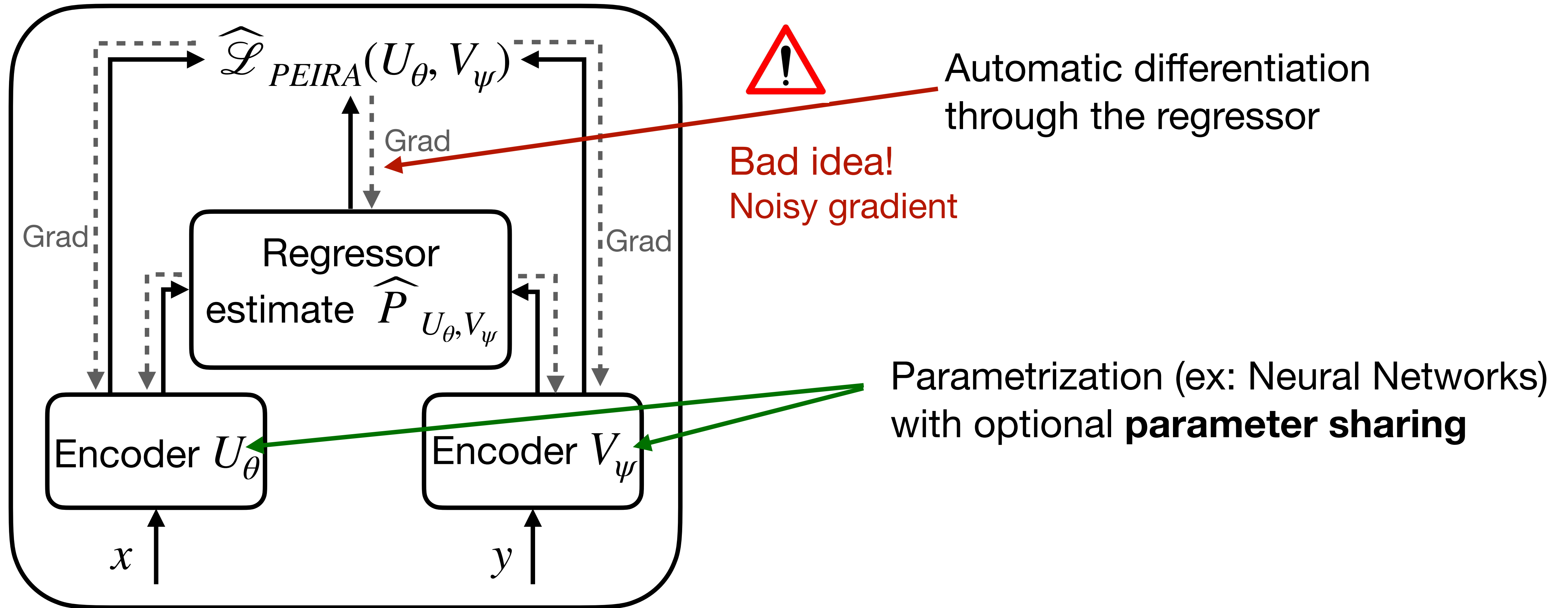
PEIRA objective: Stability and non-collapse



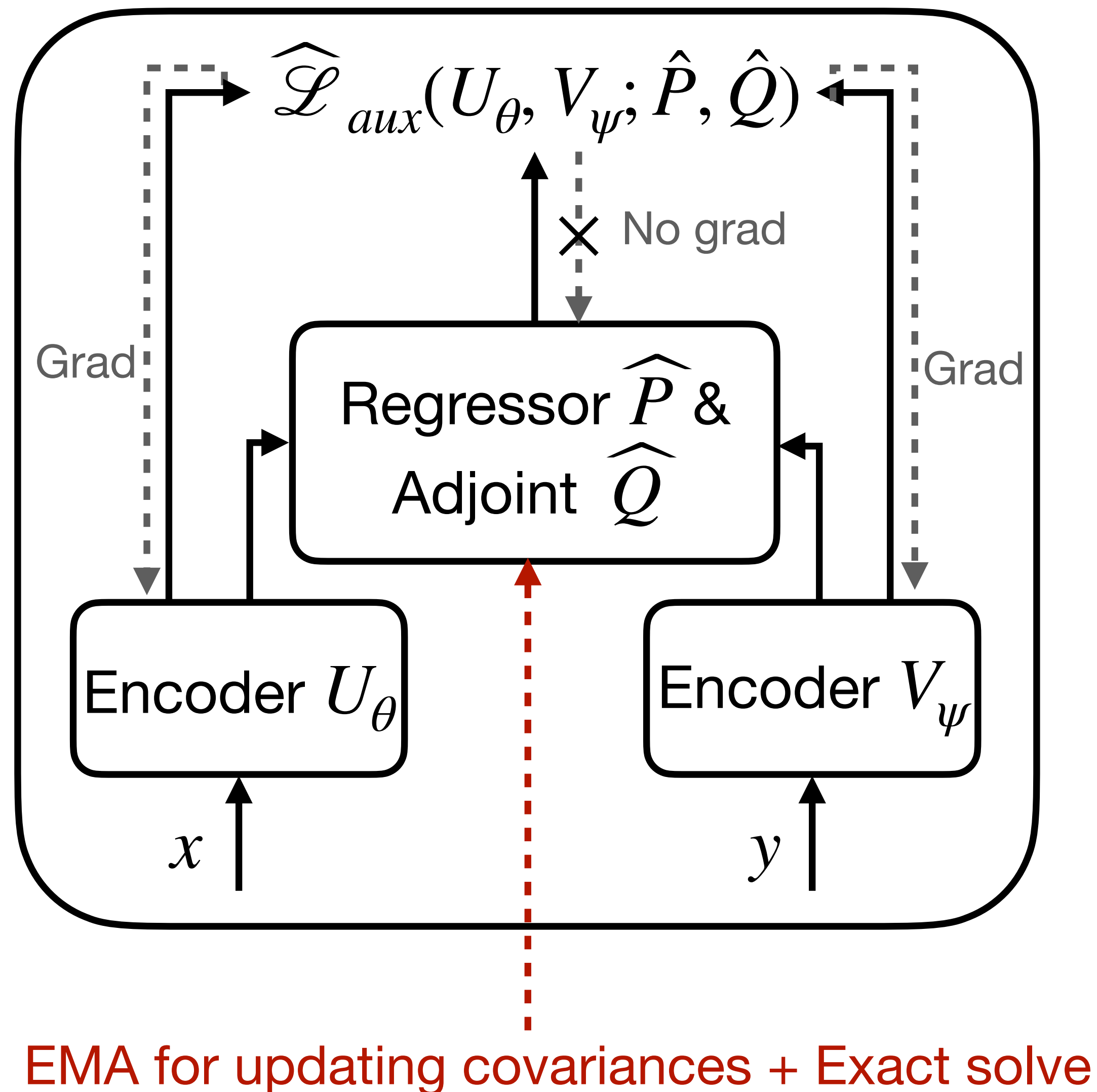
Theorem (Informal): Under the same assumptions as before:

- 1- Only global minimizers are stable.
- 2- All other critical points are unstable.

Optimizing PEIRA objective: A "bad" Algorithm



Optimizing PEIRA objective: A better Algorithm



Proposition:

$\mathcal{L}_{PEIRA}(U, V)$ has the same gradient as that of an auxiliary loss $\mathcal{L}_{aux}(U, V; P, Q)$ if:

- P is the optimal regressor
- Q satisfies

$$Q = (\mathbb{E} [U(x)U(x)^\top + V(y)V(y)^\top] + \lambda I)^{-1}$$

The auxiliary loss:

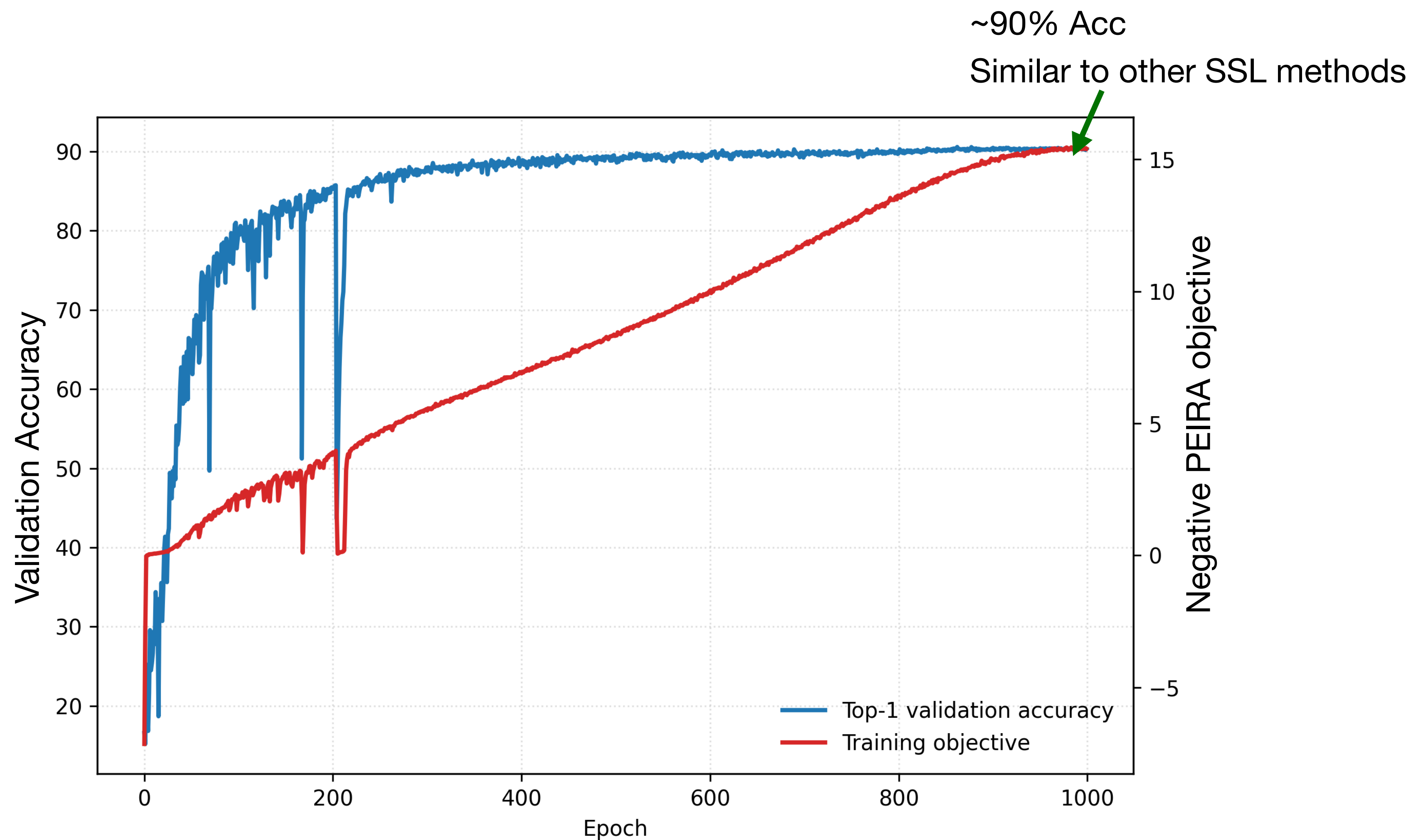
$$\begin{aligned} \mathcal{L}_{aux}(U, V; P, Q) = & \frac{1}{2} \mathbb{E} \left[QU(x)^\top (PU(x) - V(y)) \right] \\ & + \frac{1}{2} \mathbb{E} \left[QV(y)^\top (PV(y) - U(x)) \right] \\ & + \frac{\lambda}{2} \mathbb{E} [\|U(x)\|^2 + \|V(y)\|^2] \end{aligned}$$

Pretraining on ImageNet1k and Cifar10

Method	Approach	ImageNet-1K $\uparrow$	CIFAR-10 $\uparrow$
BYOL [Grill et al., 2020]	self-distillation (EMA)	69.3	–
SimSiam [Chen and He, 2021]	self-distillation (stop-gradient)	67.9	–
SimCLR [Chen et al., 2020]	contrastive (negatives)	68.5	–
VICReg [Bardes et al., 2022]	optimization	68.81 ± 0.09	90.92 ± 0.14
SIGReg [Balestriero and LeCun, 2025]	optimization	66.26 ± 0.28	92.34 ± 0.21
PEIRA (ours)	optimization	66.50 ± 0.02	90.97 ± 0.10

Competitive with some SOTA methods

SSL for images: Cifar10



Strong correlation between Val acc and PEIRA objective

Conclusion

Summary

- There is a formal connexion between some self-distillation SSL methods and non-linear CCA
- Collapse is still an issue for these methods
- PEIRA leverages this connection with CCA while provably avoiding collapse
- Promising results on image representation

Future directions/open questions

- More applications? Video, time series, world-models
- Smarter algorithms? Avoiding full matrix inversion? Variance reduction, etc
- Statistical guarantees?

Thank you!